ROSLYN WILLOUGHBY

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EDUCATION

Washington State University **Pullman, WA**
BA, Social Science *Dec 2025*

GPA: 3.75/4.00, magna cum laude
McCroskey Scholar

Research Interest: AI and Ethics
Relevant Coursework: Probability & Statistics, Python Programming, Statistics in Psychology

SKILLS

Programming & Data Processing: Python, SQL, R, Tableau, Excel
Computational Modeling and Analysis: ANOVA, Logistic Regression, Multi-objective optimization
Machine Learning: Supervised classification, model characterization and evaluation including LLMs and GNNs
Network & Behavioral Analysis: Social network analysis (Gephi), stakeholder analysis
Relevant Training: NVIDIA Fundamentals of Deep Learning, UC Davis Social Network Analysis, System Design, Presentation and Public Speaking

RESEARCH EXPERIENCE

WSU / NIH Training in Gerontechnology **Pullman, WA**
Trainee *January 2026 - February 2026*
Program led by Dr. Diane Cook & Prof. Maureen Schmitter-Edgecombe

- **Studied interdisciplinary methods from computer science and neuropsychology** that combine smart-environment and wearable sensor data with clinical cognitive assessments to detect functional decline linked to cognitive impairment earlier and more reliably.
- **Gained hands-on training in advanced gerontechnology methodologies,** including multimodal sensor fusion, longitudinal behavioral pattern modeling, and the application of machine learning techniques to aging-related health data.

USDA Research Division | WSU Phenomics Lab **Mt. Vernon, WA**
Data Science Intern *April 2024 - August 2024*
Advisor: Prof. Lisa DeVetter & Dr. Sindhuja Sankaran

- **Developed a phenomic dataset from computer-vision–extracted RGB** color metrics within a raspberry breeding project aimed at improving trait evaluation and breeder decision-making.
- **Fine-tuned a Large Language Model** (LLM) to translate quantitative phenomic outputs into breeder-facing interpretations, and characterized the efficacy of AI-generated suggestions with respect to domain expertise.
- **Evaluated model performance against expert-guided responses** and measured stakeholder confidence shifts following LLM interaction, demonstrating improved interpretability and increased trust in phenomic decision-support tools.
- **Presented findings to 50+ researchers and agricultural stakeholders**, facilitating interdisciplinary discussion on AI adoption in applied breeding systems.

PROJECT EXPERIENCE

US Movie Revenue Case Study / WSU / Probability & Statistics **Pullman, WA**
Data Analyst *December 2025*

- Analyzed financial performance data from 2006 U.S. films to test common industry claims about profitability using confidence intervals and hypothesis testing, statistically rejecting the claim that average return on investment was 12%.
- Built multiple regression models to determine which film characteristics (genre, rating, sequel status, release timing, distribution scale, critic scores) predicted opening-weekend and total box-office revenue.
- Identified key revenue drivers through model refinement, showing that number of opening theaters and sequel status predicted opening revenue, while critic ratings and opening gross explained ~80% of total domestic revenue variation.

Public Health Prediction Model **Pullman, WA**
Student Researcher *June - August 2025*
Advisor: Azam Boskabadi

- Designed and implemented an end-to-end predictive modeling pipeline, applying systematic preprocessing, feature engineering, and model validation techniques to a cervical cancer risk factors dataset in a public health context, resulting in a rigorously validated classification analysis supported by cross-validation, ROC-AUC evaluation, and diagnostic performance assessment.

WORK EXPERIENCE

WSU Access Center **Pullman, WA**
Student Supervisor *August 2023 - December 2025*

- Led recruitment and supervision of 20+ academic support specialists while coordinating accommodations for 150+ students per term, improving service quality, compliance, and timely access to resources.
- Led onboarding meetings for new student workers and built spreadsheets and checklists to track their hiring paperwork and confirm they were ready to be hired by HR.
- Remediated course materials and documents into accessible formats for students using assistive technology.
- Updated and maintained student records in the office database, which served ~2,000 students.

HONORS AND AWARDS

President's Honor Roll | Washington State University 2024, 2025
Delegate, Harvard Project for Asian and International Relations, Harvard University, Cambridge, MA 2024

ABSTRACTS AND PRESENTATIONS

1. Willoughby, R. Phenomics Tools in Raspberry Breeding: Color Perception and AI-Driven Strategies for Enhancing Breeder Adoption. *REEU Research Symposium*, Washington State University, Pullman, WA. August 1, 2024. Oral Presentation.
2. Willoughby, R. Operationalizing Equity: Data-Driven Strategies in Disability and Accessibility Services. *WSU Access Center,* Washington State University, Pullman, WA. April 2025. Oral Presentation.